

PCS224 - Solid State Physics - Useful Formulas

$$\vec{F} = q\vec{E}$$

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$|\vec{F}| = |q||\vec{v}||\vec{B}| \sin \theta$$

$$|\vec{E}|_{\text{plate}} = \frac{Q}{2A\epsilon_0} = \frac{\sigma}{2\epsilon_0}$$

$$K = mv^2/2$$

$$\Delta K + \Delta U = 0$$

$$\Delta U = q\Delta V$$

$$\Delta V = -\vec{E} \cdot \Delta \vec{x} \text{ or } -\vec{E} \cdot \Delta \vec{y}$$

$$|\Delta V| = |\vec{E}||\Delta x| \text{ or } |\vec{E}||\Delta y|$$

$$\Delta V = IR$$

$$P = I\Delta V = I^2 R = \frac{\Delta V^2}{R}$$

$$f\lambda = c$$

$$E = hf = hc/\lambda$$

$$hf = K_{\text{max}} + \phi$$

$$K_{\text{max}} = eV_{\text{stop}}$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = nqAv_d$$

$$J = I/A$$

$$\vec{J} = \sigma \vec{E}$$

$$\vec{v}_{\text{dp}} = \mu_p \vec{E}$$

$$\vec{v}_{\text{dn}} = -\mu_n \vec{E}$$

$$\sigma = \frac{1}{\rho} = e[n\mu_n + p\mu_p]$$

$$R = \rho L/A = L/(\sigma A)$$

$$\Delta V_H = \frac{IB}{nqt_B} \text{ or } \Delta V_H = \frac{IB}{nqd}$$

$$E_n = \frac{-13.6\text{eV}}{n^2}$$

$$N_C = \frac{2.5 \times 10^{19}}{\text{cm}^3} \left(\frac{m_n^*}{m_e}\right)^{3/2} \left(\frac{T}{300 \text{ K}}\right)^{3/2}$$

$$N_C(T) = N_C(300 \text{ K}) \cdot \left(\frac{T}{300 \text{ K}}\right)^{3/2}$$

$$N_V = \frac{2.5 \times 10^{19}}{\text{cm}^3} \left(\frac{m_p^*}{m_e}\right)^{3/2} \left(\frac{T}{300 \text{ K}}\right)^{3/2}$$

$$N_V(T) = N_V(300 \text{ K}) \cdot \left(\frac{T}{300 \text{ K}}\right)^{3/2}$$

$$n_0 = N_C e^{\frac{-(E_C - E_F)}{k_B T}} = n_i e^{\frac{E_F - E_{Fi}}{k_B T}}$$

$$p_0 = N_V e^{\frac{-(E_F - E_V)}{k_B T}} = n_i e^{\frac{E_{Fi} - E_F}{k_B T}}$$

$$n_0 p_0 = n_i^2$$

$$n_i^2 = N_C N_V e^{\frac{-E_g}{k_B T}}$$

$$N_d - N_a = n_0 - p_0$$

$$E_C - E_F = k_B T \ln \left(\frac{N_C}{n_0}\right)$$

$$E_F - E_V = k_B T \ln \left(\frac{N_V}{p_0}\right)$$

$$E_F = E_{Fi} + \frac{k_B T}{2} \ln \left(\frac{n_0}{p_0}\right)$$

$$E_{Fi} = E_{\text{mid-gap}} + \frac{3k_B T}{4} \ln \left(\frac{m_p^*}{m_n^*}\right)$$

$$f_F(E) = \frac{1}{1 + e^{\frac{E - E_F}{k_B T}}}$$

$$f_H(E) = 1 - f_F(E) = \frac{1}{1 + e^{\frac{E_F - E}{k_B T}}}$$

$$V_{\text{bi}} = \frac{k_B T}{e} \ln \left(\frac{N_a N_d}{n_i^2}\right)$$

$$|\vec{E}|_{\text{max}} = \frac{eN_d x_n}{\epsilon_s} = \frac{eN_a x_p}{\epsilon_s} = \frac{2(V_{\text{bi}} + V_R)}{W}$$

$$\text{where } \epsilon_s = \kappa \epsilon_0$$

$$x_n N_d = x_p N_A$$

$$x_n = \sqrt{\frac{2\epsilon_s(V_{bi}+V_R)}{e} \left(\frac{N_A}{N_d}\right) \left(\frac{1}{N_A+N_d}\right)}$$

$$x_p = \sqrt{\frac{2\epsilon_s(V_{bi}+V_R)}{e} \left(\frac{N_d}{N_A}\right) \left(\frac{1}{N_A+N_d}\right)}$$

$$W = \left[\frac{2\epsilon_s(V_{bi}+V_R)}{e} \left(\frac{N_A+N_d}{N_A N_d}\right) \right]^{1/2}$$

$$C = \frac{Q}{\Delta V} = \frac{dQ}{dV} = \frac{\epsilon_s A}{d} = \frac{\epsilon_s A}{W}$$

$$C' = C/A$$

$$C'_{ox} = \epsilon_{ox}/t_{ox}$$

$$Q_s = \tau_t I_D$$

$$Q' = Q/A$$

$$I_D = I_S [e^{eV/k_B T} - 1], I_S \propto \exp\left(\frac{-E_g}{k_B T}\right)$$

$$I_{tot} = I_L - I_D$$

$$V_{OC} = \frac{k_B T}{e} \ln \left[1 + \frac{I_L}{I_S} \right]$$

$$e\phi_{Fp} = E_F - E_{Fi, bulk} \approx -k_B T \ln \left[\frac{N_A}{n_i} \right]$$

$$e\phi_{Fn} = E_F - E_{Fi, bulk} \approx k_B T \ln \left[\frac{N_d}{n_i} \right]$$

$$e\phi_s = E_{Fi, bulk} - E_{Fi, surface}$$

$$\phi_{sT} = 2|\phi_{Fp}| = \mathbf{2\phi_{Fn}}$$

$$|Q_{SD}| = eN_A x_d A \text{ or } eN_d x_d A$$

$$x_d = \sqrt{\frac{2\epsilon_s |\phi_s|}{eN_a}} \text{ or } \sqrt{\frac{2\epsilon_s |\phi_s|}{eN_d}}$$

$$e\phi_{ms} = E_{F, semi} - E_{F, metal}$$

$$V_{FB} = \phi_{ms} - \frac{Q_{SS}}{C_{ox}} = \phi_{ms} - \frac{Q'_{SS} t_{ox}}{\epsilon_{ox}}$$

$$V_G = \phi_{ms} + V_{ox} + \phi_s$$

$$V_{TN} = \phi_{ms} + \frac{t_{ox}(|Q'_{SD, max}| - Q'_{SS})}{\epsilon_{ox}} + 2|\phi_{Fp}|$$

$$V_{TP} = \phi_{ms} - \frac{t_{ox}(|Q'_{SD, max}| + Q'_{SS})}{\epsilon_{ox}} - 2\phi_{Fn}$$

$$K_n = \frac{\mu_n W \epsilon_{ox}}{2L t_{ox}} \quad K_p = \frac{\mu_p W \epsilon_{ox}}{2L t_{ox}}$$

$$I_D = K_n [2(V_{GS} - V_{TN}) V_{DS} - V_{DS}^2]$$

$$I_D = K_p [2(V_{SG} + V_{TP}) V_{SD} - V_{SD}^2]$$

$$V_{DS, sat} = V_{GS} - V_{TN}$$

$$V_{SD, sat} = V_{SG} + V_{TP}$$

$$I_{D, sat} = K_n (V_{GS} - V_{TN})^2$$

OR

$$I_{D, sat} = K_p (V_{SG} + V_{TP})^2$$

$$|Q'_{acc}| = C'_{ox} |V_{GS} - V_{FB}| \text{ or } Q'_{acc} = -C'_{ox} (V_{GS} - V_{FB})$$

$$|Q'_{inv}| = C'_{ox} (V_{GS} - V_{TN}) \text{ or } C'_{ox} (V_{SG} + V_{TP})$$

OR

$$Q'_{inv} = -C'_{ox} (V_{GS} - V_T)$$

$$x_{DB} = \sqrt{\frac{\epsilon_s k_B T}{e^2 N_a}}$$

$$\epsilon_{ox} = 3.9$$

$$C'(\text{depl}) = \frac{\epsilon_{ox}}{t_{ox} + \frac{\epsilon_{ox}}{\epsilon_s} x_d}$$

FUNDAMENTAL CONSTANTS AND UNITS

$$e = 1.6 \times 10^{-19} \text{C}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$h = 6.64 \times 10^{-34} \text{ J} \cdot \text{sec}$$

$$c = 3.0 \times 10^8 \frac{\text{m}}{\text{sec}}$$

$$hc = 1240 \text{eV} \cdot \text{nm}$$

$$k_B = 1.38 \times 10^{-23} \text{ J/K} = 8.617 \times 10^{-5} \text{eV/K}$$

$$k_B T = 0.026 \text{eV} \left(\frac{T}{300 \text{ K}} \right)$$

$$k_c = 9.0 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{F}}{\text{m}} = 552000 \frac{\text{e}}{\text{V} \cdot \text{cm}}$$

The dielectric constant of silicon dioxide $\epsilon_{ox} = 3.9\epsilon_0$

$$a_0 = 0.53 \times 10^{-10} \text{ m}$$

$$1 \text{eV} = 1.6 \times 10^{-19} \text{ J}$$

$$1 \text{ N} = 1 \text{ kg} \frac{\text{m}}{\text{sec}^2}$$

$$1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ kg} \frac{\text{m}^2}{\text{sec}^2}$$

$$1 \text{ W} = 1 \text{ J/s}$$

$$1 \text{ V} = 1 \text{ J/C}$$

$$1 \text{ A} = 1 \text{ C/sec}$$

$$1 \text{ T} = 1 \text{ kg}/(\text{C} \cdot \text{sec}) = 1 \text{ V} \cdot \text{sec}/\text{m}^2$$

$$1 \text{ F} = 1 \text{ C/V}$$

$$1 \text{\AA} = 10^{-8} \text{ cm}$$

Table B.4 | Silicon, gallium arsenide, and germanium properties ($T = 300\text{ K}$)

Property	Si	GaAs	Ge
Atoms (cm^{-3})	5.0×10^{22}	4.42×10^{22}	4.42×10^{22}
Atomic weight	28.09	144.63	72.60
Crystal structure	Diamond	Zincblende	Diamond
Density (g/cm^{-3})	2.33	5.32	5.33
Lattice constant ($\text{\AA}$)	5.43	5.65	5.65
Melting point ($^{\circ}\text{C}$)	1415	1238	937
Dielectric constant	11.7	13.1	16.0
Bandgap energy (eV)	1.12	1.42	0.66
Electron affinity, χ (V)	4.01	4.07	4.13
Effective density of states in conduction band, N_c (cm^{-3})	2.8×10^{19}	4.7×10^{17}	1.04×10^{19}
Effective density of states in valence band, N_v (cm^{-3})	1.04×10^{19}	7.0×10^{18}	6.0×10^{18}
Intrinsic carrier concentration (cm^{-3})	1.5×10^{10}	1.8×10^6	2.4×10^{13}
Mobility ($\text{cm}^2/\text{V}\cdot\text{s}$)			
Electron, μ_n	1350	8500	3900
Hole, μ_p	480	400	1900
Effective mass $\left(\frac{m^*}{m_0}\right)$			
Electrons	$m_I^* = 0.98$ $m_t^* = 0.19$	0.067	1.64 0.082
Holes	$m_{th}^* = 0.16$ $m_{hh}^* = 0.49$	0.082 0.45	0.044 0.28
Effective mass (density of states)			
Electrons $\left(\frac{m_n^*}{m_0}\right)$	1.08	0.067	0.55
Holes $\left(\frac{m_p^*}{m_0}\right)$	0.56	0.48	0.37

